**The Unduscular Equations**

For the electron equation, we start from the Unduscular theory ("onde" and "corpuscle" in French), is referred to an elementary particle with no mass and which moves at the speed of light and acts like a wave and a particle: photon, graviton, magnetron, particle$_T$, etc., and a particle that has mass and a speed smaller than light (atomic nucleus, positron, electron, particle$_M$…). The unduscular equation $\frac{2m}{h^2}(E-U(r))f(r,t)+\frac{R}{r}\frac{\partial^2(f(r,t))}{\partial r^2}=\frac{1}{V^2}\frac{\partial^2 f(r,t)}{\partial t^2}$ (1) is valid in both spherical and cylindrical coordinates; it emerges from a combination of the Schrödinger equation, which is the first degree with respect to time, and the Klein-Gordon temporal equation of the second degree with respect to time. Also, we chose a parameter R; thus, for a small radius $r_{Bohr,}$ for an atomic value coincides with the Schrödinger equation exactly and predicts the evolution of the wave function to huge values of r, about the galactic distances. The wave function of the Schrödinger equation has the supposition that it is null at infinity, is normed, has the dimension of $[L^{-3/2}]$, and predicts the probability of the existence of the microparticle in volume dV. Schrödinger's equation is (2):

$\frac{2m}{\hbar^2}(E-U(r))f(r,t)+\frac{\partial^2(f(r,t))}{\partial r^2}=ih\frac{\partial f(r,t)}{\partial t}$ & Klein-Gordon is $\frac{2m}{\hbar^2}f(r,t)+\frac{\partial^2(f(r,t))}{\partial r^2}=\frac{1}{c^2}\frac{\partial^2 f(r,t)}{\partial t^2}$ (3) Eq. (1) for m = 0 is a wave equation with variable parameters and seems to be true. $\frac{R}{r}\frac{\partial^2(rf(r,t))}{\partial r^2}=\frac{1}{V^2}\frac{\partial^2 rf(r,t)}{\partial t^2}$ (4), but all 3 equations are different, and the wave equation is not satisfactory $\frac{\partial^2(f(r,t))}{\partial r^2}=\frac{1}{c^2}\frac{\partial^2 f(r,t)}{\partial t^2}$ (5),

$\frac{2m}{\hbar^2}(E-U(r))rf1(r,t)+\frac{R\partial^2(rf1(r,t))}{r\partial r^2}=\frac{\partial^2 rf1(r,t)}{V^2\partial t^2}$ and f1(r,t)= g(r)u(t) thus

$\frac{2m}{\hbar^2}\big(E-U(r)\big)rg(r)u(t)+\frac{R\partial^2(rg(r)u(t))}{r\partial r^2}=\frac{\partial^2(rg(r)u(t))}{V^2\partial t^2}$

$\frac{2m}{\hbar^2}\big(E-U(r)\big)rg(r)u(t)+u(t)\frac{R\partial^2 rg(r)}{r\partial r^2}=\frac{rg(r)\partial^2 u(t)}{V^2\partial t^2}$ and separating variables

$\frac{2m}{\hbar^2}(E-U(r))+\frac{R}{g(r)r^2}\frac{\partial^2(r\cdot g(r))}{\partial r^2}=\frac{1}{u(t)\cdot V^2}\frac{\partial^2 u(t)}{\partial t^2}$ thus $\frac{1}{u(t)\cdot V^2}\frac{\partial^2 u(t)}{\partial t^2}=\ k^2$ with the sinusoidal solution.

$\frac{2m}{\hbar^2}(E-U(r))+\frac{R}{g(r)r^2}\frac{\partial^2(r\cdot g(r))}{\partial r^2}=k^2$

$\frac{2m}{\hbar^2}[(E-U(r))-\frac{\hbar^2}{2m}k^2]\cdot g(r)+\frac{R}{r^2}\frac{\partial^2(r\cdot g(r))}{\partial r^2}=0$ thus

$\frac{2m}{\hbar^2}[(E-U(r))-\frac{\hbar^2}{2m}k^2]\cdot g(r)+\frac{R}{r^2}\frac{\partial^2(r\cdot g(r))}{\partial r^2}=0$ & $u(t)=\sinh(Vkt)$, so we calculate k that should be

for any small r, almost equal to Schrödinger $\frac{2m}{\hbar^2}(E-U(r))\cdot g(r)+\frac{1}{r}\frac{\partial^2(r\cdot g(r))}{\partial r^2}=0$ & $u(t)=e^{-\left(\frac{i}{\hbar}\right)Et}$

Thus we have substitute rg(r)u(t) = $f_1$(r,t) thus: $\frac{2m}{\hbar^2}(E-U(r))f_1(r,t)+\frac{R}{r}\frac{\partial^2(f_1(r,t))}{\partial r^2}=\frac{1}{V^2}\frac{\partial^2 f_1(r,t)}{\partial t^2}$

This is the equation for any micro particle that we take for granted; we obtain equation (1) also for v = c; thus, m = 0, we have equation (5). See: https://michaelvio.github.io/Supposition.pdf

### 1.1 Step 1 — Energy balance for a bound electron

For a hydrogenic electron in a Coulomb field, the non-relativistic energy is $E=\frac{1}{2}m_eV^2-\frac{e^2}{4\pi\varepsilon_0 r}+\frac{l(l+1)\hbar^2}{2m_e r^2}$. (1.1)

Solving for the kinetic term and dividing by $\hbar^2/(2m_e)$ gives the "wavenumber-squared" combination

$$\frac{2m_e}{\hbar^2}\left[E+\frac{e^2}{4\pi\varepsilon_0 r}-\frac{l(l+1)\hbar^2}{2m_e r^2}\right]=\frac{m_e^2V^2}{\hbar^2}=k^2(r). \qquad (1.2)$$

### 1.2 Step 2 — Onduscular wave ansatz

In TauTimp, an electron is represented by a spatial-temporal wave $f(r,t)$ propagating at the local "ondoscopic" speed $V$, with the cosmological scale $R$ entering through a curvature-corrected Laplacian. The local d'Alembertian satisfies

$$\Box f = \frac{R}{r^2}\frac{\partial^2(rf)}{\partial r^2} - \frac{1}{V^2}\frac{\partial^2 f}{\partial t^2} = 0. \qquad (1.3)$$

The prefactor $R/r^2$ encodes the cosmological coupling: at $r = R$ the operator reduces to the standard $\partial_r^2$ on the conformal coordinate, while at atomic scale $r \ll R$ it amplifies the radial derivatives by $R/r^2 \sim 10^{22+22} = 10^{44}$, which is exactly the factor needed to make $E/\hbar$ and $k^2(r)$ commensurable in the combined equation below.

### 1.3 Step 3 — Coupling the wave equation to the energy condition

Combining (1.2) and (1.3) and adding the local Coulomb + centrifugal source on the LHS:

$$\boxed{\frac{2m_e}{\hbar^2}\left[E + \frac{e^2}{4\pi\varepsilon_0 r} - \frac{l(l+1)\hbar^2}{2m_e r^2}\right] f(r,t) \; + \; \frac{R}{r^2}\frac{\partial^2(r\,f(r,t))}{\partial r^2} \; = \; \frac{1}{V^2}\frac{\partial^2 f(r,t)}{\partial t^2}.} \qquad (1)$$

This is the **onduscular (TauTimp) electron equation**, in its raw form before separation of variables. With the cosmological radius $R = 3.9228 \times 10^{22}$ m and the boundary condition $g(R) = 0$ (where $f(r,t) = g(r)\,u(t)$), it determines the bound spectrum.

### 1.4 Step 4 — Separation of variables

Let $f(r,t) = g(r)\,u(t)$ with $u(t) = \cos(\omega t)$, so that $\partial_t^2 u/u = -\omega^2$. The energy condition $E = \hbar\omega$ relates the temporal frequency to the electron energy. Substituting into (1):

$$\frac{2m_e}{\hbar^2}\left[E + \frac{e^2}{4\pi\varepsilon_0 r} - \frac{l(l+1)\hbar^2}{2m_e r^2}\right] g(r) + \frac{R}{r^2}\frac{d^2(r\,g(r))}{dr^2} = -\frac{\omega^2}{V^2}\,g(r).$$

The RHS contributes a term $-\omega^2/V^2 \cdot g$ that combines with the energy term on the LHS. For non-relativistic bound states, $V^2 = 2E/m_e$, so $\omega^2/V^2 = E^2 m_e/(2\hbar^2 E) = E m_e/(2\hbar^2)$, which is already absorbed into the $2m_e E/\hbar^2$ term. The radial equation becomes $\frac{R}{r^2}\frac{d^2(r\,g(r))}{dr^2} + g(r)\left[\frac{2m_e E}{\hbar^2} + \frac{2}{a_0\,r} - \frac{l(l+1)}{r^2}\right] = 0, \qquad (2)$

where we used $2m_e e^2/(\hbar^2 \cdot 4\pi\varepsilon_0) = 2/a_0$.

This is the **original radial ODE TauTimp** used for the Taylor comparison.

The complete onduscular equation, in the form that reproduces the entire atomic structure of hydrogen, is

$$\boxed{\begin{aligned}&\frac{2m_e}{\hbar^2}\left[E + \frac{e^2}{4\pi\varepsilon_0\,r} - \frac{l(l+1)\hbar^2}{2m_e r^2} + \xi(r)\,\langle\vec{L}\cdot\vec{S}\rangle + V_{HF}(r)\,\langle\vec{S}_e\cdot\vec{I}_p\rangle\right] f(r,t)\\ &\qquad + \frac{R}{r^2}\frac{\partial^2(r\,f(r,t))}{\partial r^2} \; = \; \frac{1}{V^2}\frac{\partial^2 f(r,t)}{\partial t^2}.\end{aligned}} \qquad (3)$$

with - $\xi(r) = \frac{1}{2m_e^2 c^2\,r}\frac{dV}{dr} = \frac{\hbar\alpha}{2m_e^2 c}\frac{1}{r^3}$ for Coulomb $V(r) = -\hbar c\alpha/r$, - $V_{HF}(r) = \frac{2\mu_0}{3}\,g_p\,\mu_B\,\mu_N\,\delta^3(\vec{r}) \cdot \frac{4}{\hbar^2}$ — the Fermi electron–proton contact term.

Equation (3) is the centerpiece of this document.

## 2. The two governing equations (for comparison with Schrödinger) 2.1 Schrödinger radial equation (hydrogen, reduced form $u = r\psi$)

$$\mathcal{L}_S\,u(r) \;\equiv\; \frac{d^2u}{dr^2} + u(r)\left[\frac{2m_e E}{\hbar^2} + \frac{2}{a_0 r} - \frac{l(l+1)}{r^2}\right] \;=\; 0. \qquad (4)$$

### 2.2 ODE TauTimp (original operator, scalar, no spin)

From (2):

$$\frac{R}{r^2}\frac{d^2(r\,g(r))}{dr^2} + g(r)\left[\frac{2m_eE}{\hbar^2} + \frac{2}{a_0\,r} - \frac{l(l+1)}{r^2}\right] = 0. \qquad (5)$$

With $u(r) = r\,g(r)$, the kinetic term becomes $(R/r^2)\,u''(r)$. Multiplying by $r^2/R$ to clear singularities:

$$u''(r) + \frac{u(r)}{R}\cdot r\left[\frac{2m_eE}{\hbar^2} + \frac{2}{a_0 r} - \frac{l(l+1)}{r^2}\right] = 0.$$

This is the form used in the Taylor comparison below.

### 3. Taylor comparison at $r = 0$

Multiply (4) by $r^2$ and (5) (in the $u = rg$ form) by $r^2/R$ to clear $1/r$ and $1/r^2$. Substitute $u(r) = \sum_{n\geq 0} u_n\, r^n$ with the regularity boundary condition $u_0 = 0$ (so that $g(r) = u(r)/r$ is finite at the origin).

### 3.1 Coefficient table (at $l = 0$, regular solution $u_0 = 0$)

| $N$ | $[r^N]$ of Schrödinger $\times\, r^2$ | $[r^N]$ of TauTimp $\times\, r^2/R$ |
|---|---|---|
| 0 | 0 (since $u_0 = 0$) | 0 |
| 1 | $2u_0/a_0 = 0$ | $2u_0/(a_0R)\cdot R = 0$ |
| 2 | $2u_2 + 2u_1/a_0 + \frac{2m_eE}{\hbar^2}u_0$ | $\frac{1}{R}[2u_2 + 2u_1/a_0]$ |
| 3 | $6u_3 + 2u_2/a_0 + \frac{2m_eE}{\hbar^2}u_1$ | $\frac{1}{R}[6u_3 + 2u_2/a_0]$ |
| ⋮ | ⋮ | ⋮ |
| 6 | $30u_6 + 2u_5/a_0 + \frac{2m_eE}{\hbar^2}u_4$ | $\frac{1}{R}\left[30u_6 + 2u_5/a_0 + \frac{2m_eE}{\hbar^2}u_0\right]$ |

### 3.2 Conclusion of the Taylor comparison

- $r^0$ **coefficient**: equal (both zero) — guaranteed by the regularity condition $u_0 = 0$. ✓
- $r^1, r^2$: equal trivially because $u_0 = 0$.
- **Divergence at** $r^3$: Schrödinger has the energy term $\frac{2m_eE}{\hbar^2}u_1$; TauTimp does not (it appears only at $r^6$ and is additionally suppressed by $1/R$).

**Net effect**: the original TauTimp ODE misses the energy term $\frac{2m_eE}{\hbar^2}$ at atomic scale; the centrifugal term carries $1/R$ instead of $1/r^2$.

### 4. Born's rule $P \sim |\Psi|^2$ and the Madelung density 4.1 Starting point

Above (and in suppositions 6 and 7 below) I state that every microparticle has the probability density $p(x,t) = |\psi(x,t)|^2$, "as in Bohm's Quantum Potential". In standard quantum mechanics this is a postulate (Born, 1926). This chapter shows how the rule can be *motivated* for the onduscular equation (1)–(3), and says exactly what is demonstrated and what remains a hypothesis.

One point must be settled first. The onduscular equation is real and of second order in time, while Born's rule is written for a complex wave function. The separated solution $f(r,t) = g(r)\cos(\omega t)$ is the real part of the complex mode

$$\Psi(r,t) = g(r)\,e^{-i\omega t}, \qquad f(r,t) = \mathrm{Re}\Psi(r,t). \qquad (4.1)$$

Both carry the same information, because $u(t) = \cos\omega t$ is fixed by $E = \hbar\omega$. The time average of the real function over one period gives the same density, up to a constant factor that the normalization removes:

$$\overline{f^2} = \frac{1}{T}\int_0^T g^2(r)\cos^2(\omega t)\,dt = \frac{1}{2}\,g^2(r) = \frac{1}{2}\,|\Psi|^2. \qquad (4.2)$$

So for the onduscular equation, $P \sim |\Psi|^2$ means $P \sim g^2(r)$: the probability follows the time-averaged square of the real onduscular wave.

### 4.2 Step 1: the substrate is a fluid, and $|\Psi|^2$ is its density (Madelung)

Write the wave function in polar form, $\Psi = \sqrt{\rho}\, e^{iS/\hbar}$. The Schrödinger equation splits exactly into two fluid equations (Madelung, 1927):

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho\, \vec{v}) = 0, \qquad \vec{v} = \frac{\nabla S}{m}, \qquad (4.3)$$

$$m\left(\frac{\partial \vec{v}}{\partial t} + (\vec{v} \cdot \nabla)\vec{v}\right) = -\nabla(U + Q), \qquad Q = -\frac{\hbar^2}{2m}\frac{\nabla^2\sqrt{\rho}}{\sqrt{\rho}}. \qquad (4.4)$$

Equation (4.3) is the **continuity equation** for the density $\rho = |\Psi|^2$. Equation (4.4) is Euler's equation for a fluid with one extra term, Bohm's **quantum potential** $Q$. This is the Bohm quantum potential that I refer to in the suppositions, so the density that enters Bohm's theory is exactly the Madelung density $\rho = |\Psi|^2$.

**Link with the continuity equation used in my papers.** The transport equation I cite from Brian Greene and from the UCSD notes,

$$\frac{\partial p(x,t)}{\partial t} + \frac{\partial (p(x,t)\, V(x))}{\partial x} = 0,$$

is equation (4.3) in one dimension with the fluid velocity $v = V$. The first-order equation of the system in the propagation paragraph, $\frac{1}{V}\frac{\partial P}{\partial t} + \frac{\partial P}{\partial r} = 0$, is the same equation for constant $V$. Its general solution is $P(r,t) = F(r - Vt)$: the probability is carried unchanged at the speed $V$. So the probability wave of the onduscular theory *is* a Madelung density.

**Radial form for the onduscular wave function.** For the bound state (4.1), $S = -\hbar\omega t = -Et$ does not depend on $r$, so $\vec{v} = 0$ and $\rho = g^2(r)$ is stationary. This is the known Bohm result for real stationary states: the density stays in place, and the quantum potential exactly balances the Coulomb force, $\nabla(U + Q) = 0$. The probability of finding the particle in the shell $[r, r + dr]$ is

$$dP = |g(r)|^2\, 4\pi r^2\, dr = 4\pi\, f_1{}^2(r)\, dr, \qquad f_1(r) = r\, g(r). \qquad (4.5)$$

Here $f_1 = r\, g$ is the Effective Onduscular Wave Function (EOWF). The dimensions agree with those given for the OWF and EOWF: $g$ has $[L^{-3/2}]$, so $g^2 r^2\, dr$ is dimensionless; $f_1$ has $[L^{-1/2}]$, so $f_1{}^2$ is a probability per unit length. The normalization with the onduscular boundary condition $g(R) = 0$ (or $g(2R) = 0$) is

$$4\pi \int_0^R f_1{}^2\,(r)\, dr = 1. \qquad (4.6)$$

The EOWF is therefore the natural function for Born's rule: its square is directly the radial probability density.

**Bonus: quantized circulation.** Because $\Psi$ has a single value at every point, the circulation of $\vec{v}$ around a vortex line is quantized:

$$\oint \vec{v} \cdot d\vec{l} = n\,\frac{2\pi\hbar}{m} = n\,\frac{h}{m}, \qquad n \in \mathbb{Z}. \qquad (4.7)$$

This is the winding-number condition used for the quantization of angular momentum. In superfluids (helium, Bose–Einstein condensates) it is observed directly (Onsager, Feynman).

### 4.3 Step 2: the trembling motion is the diffusion that generates the Schrödinger equation (Nelson)

Nelson (1966) showed that if a particle performs a Brownian motion with the diffusion coefficient

$$D = \frac{\hbar}{2m}, \qquad (4.8)$$

then its probability density $\rho$ and mean velocity satisfy exactly the Schrödinger equation, with $\rho = |\Psi|^2$.

**Link with the quantum jumps of the onduscular theory.** In the suppositions, the particle executes "an instant quantum jump from the successive position in space" once per time quantum. A random walk with step $\ell$ and time step $\tau$ has $D = \ell^2/(2\tau)$. Two versions give exactly Nelson's value:

- *Zitterbewegung.* Step equal to the trembling radius $r_Z = \hbar/(2mc)$ at speed $c$: $D \sim c\, r_Z = \hbar/(2m)$. Exact.
- *Bohr-orbit jumps.* Step equal to the Bohr radius $a_0$, made at the electron's velocity on the first orbital, $V = \alpha c$ (the $V$ of the onduscular equation for the 1s state):

$$D = \frac{a_0\, V}{2} = \frac{a_0\, \alpha\, c}{2} = \frac{\hbar}{2m_e}, \qquad \text{because } a_0 = \frac{\hbar}{m_e c\, \alpha}. \quad \text{(exact)} \qquad (4.9)$$

Numerically $D_e = 5.79 \times 10^{-5}$ m$^2$/s for the electron and $D_p = 3.15 \times 10^{-8}$ m$^2$/s for the proton.

If the same jump of $a_0$ is made at the speed of light, in the photon time quantum $T_q = a_0/c = 1.765 \times 10^{-19}$ s, the result is $a_0 c/2 = 7.93 \times 10^{-3}$ m$^2$/s, larger than $D_e$ by exactly $1/\alpha = 137$. Jumps of one space quantum ($SQ = 4.25 \times 10^{-19}$ m) in the primordial time quantum $PTQ = SQ/c$ give $SQ \cdot c/2 = 6.4 \times 10^{-11}$ m$^2$/s, about $10^6$ times smaller than $D_e$. So only the jumps at the particle's own speed $V$ have the right strength.

The conclusion: **the jumps have exactly the intensity that stochastic mechanics needs to produce quantum mechanics** when they are made at the particle's speed $V$. No separately tuned external noise is needed, because the mass fixes $D$. That these jumps behave as random (Brownian) noise is a **hypothesis**.

### 4.4 Step 3: equivariance and relaxation toward $|\Psi|^2$

- **Equivariance (demonstrated).** From the continuity equation (4.3): if an ensemble of particles is distributed as $|\Psi|^2$ at one moment, it stays distributed as $|\Psi(t)|^2$ at every later moment. The Born distribution is **conserved** by the dynamics.
- **For the second-order onduscular equation.** An equation of second order in time, like Klein–Gordon, conserves the density $\rho_{KG} \propto \frac{i}{2V^2}(\Psi^*\, \partial_t \Psi - \Psi\, \partial_t \Psi^*)$, not $|\Psi|^2$ itself. For one mode (4.1) this gives $\rho_{KG} \propto (\omega/V^2)\, |g|^2$, which is proportional to $|\Psi|^2$, so Born's rule holds for every stationary onduscular state. For a superposition of modes with different $\omega$, $\rho_{KG}$ and $|\Psi|^2$ differ in the interference terms; they coincide in the non-relativistic limit, where equation (1) reproduces Schrödinger at atomic scale.
- **Relaxation (an argument, not a general proof).** Valentini (1991) showed that, for sufficiently "mixing" dynamics, an arbitrary initial distribution $\rho \neq |\Psi|^2$ tends to $|\Psi|^2$. The process has an H-function that decreases, as in thermodynamics. Simulations (Valentini and Westman, 2005) show fast relaxation. In the onduscular picture the mixing comes from the quantum jumps: the trembling period ($4 \times 10^{-21}$ s for the electron), the atomic time quantum ($a_0/V = 2.4 \times 10^{-17}$ s) and the energy-reception period (about $6 \times 10^{-11}$ s for hydrogen at room temperature) are all much shorter than any measurement time. This would explain why **no deviation from Born's rule is ever seen**: the system is always relaxed.

### 4.5 Why the square and not another power

1) **The only conserved density.** The Schrödinger equation (and the Dirac equation) gives a continuity equation only for $|\Psi|^2 = \Psi^*\Psi$ (for Dirac, $\Psi^\dagger\Psi$, which is positive). For $|\Psi|$ or $|\Psi|^4$ there is no conservation law, so the total probability would not stay equal to 1.

2) **Energy density of a wave.** The onduscular equation is a wave equation, and the energy that a wave carries is proportional to the **square of its amplitude**. For the homogeneous part of the equation the energy density is

$$\varepsilon(r,t) = \frac{1}{2}\left[\frac{1}{V^2}\left(\frac{\partial f}{\partial t}\right)^2 + \left(\frac{\partial f}{\partial r}\right)^2\right] \propto \omega^2 g^2 \text{ (time average).} \qquad (4.10)$$

If the particle "materializes" where the wave carries the most energy, as supposed for the tau-neutrino temporal ionization (after $T_{quant}$ the particle appears "in the place with the highest probability density"), the probability follows the square of the amplitude. This is the same reason why the intensity of light is $\propto |E|^2$ and why the photon count in the double-slit and Mach–Zehnder experiments follows $|E|^2$.

3) **Gleason's theorem (1957).** In any Hilbert space of dimension ≥ 3, the only probability measure consistent with the structure of subspaces has the form $\langle\Psi|\hat{P}|\Psi\rangle$, which is quadratic. The onduscular equation is linear (superposition holds), so its solutions form such a space and the theorem applies.

### 4.6 Summary: what is derived and what remains a hypothesis

| Element | Status |
|---|---|
| $\rho = \|\Psi\|^2$ satisfies the continuity equation | **demonstrated** (Madelung) |
| The transport equation of my papers is the 1D Madelung continuity equation | **identity** |
| Radial probability $dP = 4\pi f_1^2\, dr$ with the EOWF $f_1 = rg$ | **demonstrated** (dimensions agree) |
| Real onduscular wave: $\overline{f^2} = 1/2\,\|\Psi\|^2$ | **demonstrated** (time average) |
| Quantized circulation $\oint v\, dl = nh/m$ | **demonstrated** (single-valued Ψ) |
| $D = a_0 V/2 = \hbar/2m_e$ for jumps at $V = \alpha c$ | **exact identity**; that the jumps are Brownian noise is a **hypothesis** |
| Jumps at speed $c$ (in $T_q$ or $PTQ$) give the Nelson $D$ | **no** — off by $1/\alpha$ or by $10^6$ |
| The $\|\Psi\|^2$ distribution is conserved (equivariance) | **demonstrated** for stationary states and in the Schrödinger limit |
| Any initial distribution relaxes to $\|\Psi\|^2$ | **argument + simulations**, not proven in general |
| Born's rule for **spin** measurements (Stern–Gerlach) | **not derived** here; it needs a full model of the measuring device |

References for this chapter: M. Born, *Z. Phys.* 37, 863 (1926); E. Madelung, *Z. Phys.* 40, 322 (1927); D. Bohm, *Phys. Rev.* 85, 166 (1952); E. Nelson, *Phys. Rev.* 150, 1079 (1966); A. Valentini, *Phys. Lett. A* 156, 5 (1991); A. Valentini and H. Westman, *Proc. R. Soc. A* 461, 253 (2005); A. M. Gleason, *J. Math. Mech.* 6, 885 (1957).

The effective wave function of the unduscular equation has the supposition that it is null at R (or 2R ), is normed, has the dimension of $[L^{-3/2}]$, and predicts the probability of the existence of the micro particle in volume dV. Also, the condition that the derivative = R at the origin verifies the condition that the speed of interaction has the velocity V, whatever will be. The value of R = 3.922833075 $\cdot 10^{22}$ m is calculated in OndSLP.pdf [The change of variable from Schrödinger to QFT Document]

As you can see, the function $f_1(r,t) = rf(r,t) = rg(r)u(t)$ is the Unduscular Wave Function (OWF) multiplied by r, denoted f(r,t) in the case of the graviton equation and other particles that travel with the speed of light, so we call it the Effective Unduscular Wave Function or EOWF. The EOWF has the dimension of $[L^{-1/2}]$, and OWF has the dimension of $[L^{-3/2}]$ with separate variables, and g(r) has dim of $[L^{-3/2}]$; u(t) is dimensionless, as in the Schrödinger equation. Both equations, Unduscular and Schrödinger, have almost the same value of the function that depends on r, g(r) multiplied by a nondimensional exponential factor that depends on time t. The relation between f1(r,t) EOWF and f(r,t) OWF is $f_1(r,t) = rf(r,t) = rg(r)u(t)$; thus we calculus EOWF which is easier, and we return to the OWF by dividing by r the function $f_1(r,t) \Rightarrow f(r,t) = f_1(r,t)/r$. The relativistic form of the undulatory equation of one particle for the velocity of the particle V < c:

$$\frac{2m}{h^2\sqrt{1-\frac{V^2}{c^2}}}(E - U(r))f(r,t) + \frac{R}{r}\frac{\partial^2(f(r,t))}{\partial r^2} = \frac{1}{V^2}\frac{\partial^2 f(r,t)}{\partial t^2}$$

https://michaelvio.github.io/SchrodTauTimp.pdf
https://michaelvio.github.io/struyve.pdf

The unduscular wave equation is linear (the equation contains only the wave function, and its derivative is not raised to any power). Thus, if we have two solutions of it, a linear combination of the solutions for one particle should also be a solution, and that is superposition. $\frac{2m}{\hbar^2}(E - U(r))f_1(r,t) + \frac{R}{r}\frac{\partial^2(f_1(r,t))}{\partial r^2} = \frac{1}{V^2}\frac{\partial^2 f_1(r,t)}{\partial t^2}$

The wave-particle duality is generated by the fact that all atoms and particles receive energy from stars and are one-time quanta particles and one-time quanta waves of probability and have the definition $p(x,t) = |\psi(x,t)|^2$ as in Bohm's Quantum Potential, and between states executes an instant quantum jump to the successive position in space due to the granularity of time. Also suppose that force-mediating particles of these interactions are https://physics.aps.org/articles/v16/13 electronic neutrinos for gravity, which is a Majorana particle with spin quantum number s = 0, photons for light, magnetrons for magnetic

forces, and tau neutrino for time, which is a Dirac particle with spin quantum number s = -½ ... Space has a special treatment, but we barely say that the muon neutrino is the place-mediating particle, which is a Dirac particle with spin quantum number s = -½. The other elementary stable particles are the electron, proton, neutron, and phonon, which are treated as an equation in TOnduscular.pdf;

Schrödinger's and unduscular equations inference document

The initial derivative = R or other constraints, the boundary condition f(R)=0 or f(2R)=0, or another initial condition was determined by dowsing. The suppositions that I make are in the file Supposition.pdf https://michaelvio.github.io/Supposition.pdf

$$\frac{1}{V^2}\frac{u''(t)}{u(t)} = \frac{1}{\varphi(r)}\left[\frac{R}{r^2}\frac{d^2}{dr^2}(r\varphi) + \frac{2m}{\hbar^2}\left(E_0 - \frac{8\pi\varepsilon}{r}\right)\varphi\right] = -\frac{\mathrm{k}^2}{V^2}$$

$$\mathrm{f}(r,t) = \varphi(r)\cdot \mathrm{u}(t)$$

$$\frac{R}{r^2}\frac{d^2}{dr^2}(r\varphi) + \left[\frac{2m}{\hbar^2}\left(E_0 - \frac{\alpha}{r}\right) + \frac{\mathrm{k}^2}{V^2}\right]\varphi(r) = 0 \qquad \text{where } \alpha = 8\pi\varepsilon. \qquad (8)$$

Substitution $y(r) = r\,\varphi(r)$

Becoause $\frac{d^2}{dr^2}(r\varphi) = y''(r)$, equation (8) becomes after multiplying $r^2/R$:

$$y''(r) + \frac{r^2}{R}\left[\frac{2m}{\hbar^2}\left(E_0 - \frac{\alpha}{r}\right) + \frac{\mathrm{k}^2}{V^2}\right]\frac{y(r)}{r} = 0$$

We simplify:

$$y''(r) + \left[\frac{r}{R}\left(\frac{2mE_0}{\hbar^2} + \frac{\Omega^2}{V^2}\right) - \frac{1}{R}\cdot\frac{2m\alpha}{\hbar^2}\right]y(r) = 0$$

**Canonical Airy:**

$$\boxed{y''(r) + (a\,r - b)y(r) = 0}$$

With coefficients:

$$a = \frac{1}{R}\left(\frac{2mE_0}{\hbar^2} + \frac{\mathrm{k}^2}{V^2}\right)$$

$$b = \frac{1}{R}\cdot\frac{2m\alpha}{\hbar^2} = \frac{2m\cdot 8\pi\varepsilon}{\hbar^2 R} = \frac{16\pi m\varepsilon}{\hbar^2 R}$$

We have solution: $\mathrm{f}(r,t) = \sum_{n=1}^{\infty}\varphi_n(r)\cdot[A_n\cos(\mathrm{k}_n t) + B_n\sin(\mathrm{k}_n t)]$ With general solution:

$$\mathrm{f}(r,t) = \sum_{n=1}^{\infty}\frac{1}{r}\left[\mathrm{Ai}(z_n(r)) - \frac{\mathrm{Ai}(\xi_0^{(n)})}{\mathrm{Bi}(\xi_0^{(n)})}\mathrm{Bi}(z_n(r))\right]\cdot[A_n\cos(\mathrm{k}_n t) + B_n\sin(\mathrm{k}_n t)]$$

https://michaelvio.github.io/Electron1s.pdf

The Unduscular Wave Function (OWF) is the value of a particle into space-time at instant t into spherical, cylindrical, or Cartesian coords specific for every particle in particular: for the graviton is for a spatial wave function [$L^{-3/2}$] and of the function of graviton f (r,t) is [$L^{-3/2}$] ...and is related to the probability to find the graviton in the local space from 0 to 2·R (where R = $3.9228\cdot10^{22}$ m) these are unduscular wave function concrete value ("valori concrete ale functiei de unda unduscular" in Romanian language) as you can see the f is bounded function ("marginita") and we can normalize that function. Besides these values 0 to 2·R for the graviton, the wave function has improper virtual values ("Valori improprii virtuale"), and the function is not bordered in fact, for r over 2R the force of attraction changes sign, goes to infinity when r->-∞ (repulsion), and is extremely high but not ∞ at the origin. At the normalization condition that is an integral with respect to r and for t fixed, t = $t_0$, we have: $4\pi\int_0^{2R}\psi(r,t_0)^2 dr = 1$ where ψ(r,t) = f(r,t)/r.

In the graviton equation, f(r,t)= u(t)·g(r), where [L] $^{-3/2,}$ and the probability of finding the graviton in the volume dV is P(r,t) = $|\psi(r,t)^2|$dV in 3-dimensional space, as in Bohm's Quantum Potential. Under the condition that the velocity of all the particles below, except the electron, is c, we recalculate the basis-separated solution and the singular solution of the unduscular equation, each satisfying a set of three singular conditions. We mention that the below-denoted $k_1$, $k_2$, $k_3$, $k_4$, and $k_5$ are real constants, which are specific to the particle. We suppose that all the functions are separable into a single function of one variable. The theory

of everything should describe all of the characteristics of the moving particle in space and time, introducing the Effective Unduscular Wave Function, which is the value of the particle in space-time at instant t in spherical, cylindrical, or Cartesian coordinates.

**I) The unduscular equation provides an epistemic description of reality, with real values.**

**II) It is not a part of reality itself, but a mathematical tool.**

**III) Its role is to describe and correctly predict the behavior of reality, closer to reality than the Schrödinger equation with complex values, over an extended spatial domain. The unduscular equation describes reality in an epistemic way, meaning it is about our knowledge of reality, not reality itself. In other words, the equation is just a mathematical tool that makes very good predictions about what happens in reality - it has real values, but it is not Reality itself.**

**1) The Graviton-Equation**

Starting from the unduscular equation, we will particularize graviton, tau neutrino, photon, magnetron, and electron into the potential field U(r) depending on the radius. It's not important to have the exact general solution as shown; we need the structure of the general solution of the unduscular equation EOWF:

$$\frac{R}{r}\Delta f(r,t) = \frac{1}{c^2}\frac{\partial^2 f(r,t)}{\partial t^2} \qquad (1)$$

where R => is a constant distance. First, we search for the basis of solutions that have separated variables; therefore, we impose the initial conditions for the graviton, such as to sort out the singular solution of interest:

A) [f(r,t) = 0 for r=2·R for any t] EOWF f(r,t) that can be replaced with g (2·R) = 0;

B) Constraint *Newton's law for r <R/20*;

C) and that verifies the condition that the speed of interaction is the velocity of light c.

The complete solution under azimuthal symmetry is

**2) The Tau Neutrino Equation**

Starting from the unduscular equation particularized for the tau neutrino, we have:

$$\frac{R}{r}\Delta f(r,t) = \frac{1}{c^2}\frac{\partial^2 f(r,t)}{\partial t^2} \qquad (1bis)$$

where R=> is a constant distance; first, we search for the basis of solutions that have separated variables; therefore, we impose the initial conditions for the tau neutrino, such as to sort out the singular solution of interest:

A) [f(r,t) = 0 for r = R for any t] EOWF f(r,t) that can be replaced with g (R) = 0;

B) Constraint $P(r_0,0) = 0$; $r_0 \approx 11222$ m, Schwarzschild's radius of the smallest known Black Hole

C) and that verifies the condition that the speed of interaction is the velocity of light c.

The above 3 conditions imposed onto the Tau Neutrino Effective Unduscular Wave Function f(r,t) uniquely determine the basis of separated solutions of interest.

**3) The Muon Neutrino Equation**

Starting from the unduscular equation particularized for the muon neutrino, which is a Dirac neutrino with spin quantum number s = -½, we have:

$$\frac{R_s}{r}\Delta f(r,t) = \frac{1}{c^2}\frac{\partial^2 f(r,t)}{\partial t^2} \qquad (1bis)$$

Where the value $2\cdot R_s = R\cdot$ dimension nucleon/dim quark, thus is around $7.0613\cdot10^{25}$ m, thus: $2R_s = 2\cdot3.9228\cdot0.841419/0.43\cdot10^{25}$ m$=1.553292368\cdot10^{26}$ m = 16.41866 billion Light Years, so $2\cdot R_s \sim 16.41866$ billion Light-years, therefore we impose the initial conditions for the muon neutrino, such as to sort out the singular solution of interest:

A) [f(r,t) = 0 for r=2·$R_s$ for any t] EOWF f(r,t) that can be replaced with g (2·$R_s$) = 0;

B) Constraint $\int_{r_{\min}}^{2R_s} F(r)\,dr = \frac{c}{H_0}$ & $\frac{P(r,0)}{4\pi r^2} \approx$ *Hubble const.*

C) and that verifies the condition that the speed of interaction is the velocity of light c.

The above 3 conditions imposed on the Muon Neutrino Effective Unduscular Wave Function f(r,t) uniquely determine the basis of separated solutions of interest.

The similarity between the muon neutrino (μ space) and the electronic neutrino (ν graviton) equation and boundary condition was the reason that, in general relativity, the gravity field and space are the same thing to Einstein, but they are not, as you can see in the file GravBend.mw

$\frac{R}{r}\Delta f_1(r,t) = \frac{1}{c^2}\frac{\partial^2 f_1(r,t)}{\partial t^2}$ (1bis)

The ratio between $R_s$ and R, $R_s/R$=1979.8 mince that there is a transformation between the 2 equations, which means the effective Undular Wave Function is proportional but not equal. If we substitute r=r/1979.8, equations (3) and (1) are equivalent; thus, the curvature of space is approximately ~1980 times smaller than the gravitational field.

The space growth rate is the flux equation link https://michaelvio.github.io/Flux.pdf

**4) The Photon Equation**

A specific solution of the unduscular equation for the photon, where R is a constant distance, is

$$\frac{R}{r}\Delta H(r,\theta,z,t) = \frac{1}{c^2}\frac{\partial^2 H(r,\theta,z,t)}{\partial t^2}\ . \quad (5)$$

Considering that the photon is invariant with respect to $\theta$ (z-axis symmetry) in cylindrical coordinates with z-axis symmetry, we are interested in searching for the separable solutions

f(r,z,t) = $H_3(r)$ $H_4(z)$ u(t) with z-axis symmetry, which satisfies the following four singular conditions for the photon

A) [f(r,z,t) = 0 for r=R for any t and z] the EOWF f(r,z,t) that can be replaced with $H_3(R) = 0$;

B) Constraint: a Gaussian envelope in $r$, and zeros of $P(r,z,t_0)$ at $t_0$ fixed, located on a periodic lattice $r = m\,r_b$, $m \in \mathbb{Z}_{\geq 0}$.

C) and that verifies the condition that the speed of a photon is the velocity of light "c".

Since $H_4(z) = \sin(\frac{E}{h\nu}z + \varphi)$, the wave pulsation is $k = \frac{E}{h\nu}$, where E is the photon energy and ~~h~~ is the reduced Planck constant.

Assembling all contributions, the **complete reconstructed field** is

$$H(r,\theta,z,t) = f(r,z,t) = \sum_{n,m} \mathcal{N}_{nm}\ \sqrt{r}\,J_{1/3}\left(\frac{2\beta_n}{3}r^{3/2}\right)\left[C_{nm}\cos(k_{z,m}z) + D_{nm}\sin(k_{z,m}z)\right]e^{-i\omega_n t}$$

where $\phi_m$ is an arbitrary azimuthal phase and $\mathcal{T}_n^{(m)}(t) = C_n^{(m)}\cos(\omega_n^{(m)}t + \delta_n^{(m)})$ is the temporal amplitude.

Where the constant distance R is about $3.9228\cdot10^{22}$ m, which is 4.15 million light-years, the same value as in the equation's cases for the graviton, the tau neutrino, the magnetron, and the electron. Only the $R_s$ constant from the quantum space is around $7.0613\cdot10^{25}$ m (Hence the value 2·Rs = R· dimension nucleon/dim quark thus is $2\cdot3.9228\cdot0.841419/0.43\cdot10^{25}$ m=$1.553292368\cdot10^{26}$ m = 16.41866 billion light-years). Thus, 2·Rs~16.41866 billion Light-years and differs from the constant R for the magnetron, photon, electron neutrino & tau neutrino. A simple observation of the star sky shows that there exists no galaxy larger in radius greater than 2R (Diameter of 2·2·4.147 = 16.588 million light-years), simply because after that distance occurs the rejection of the star, a circumstance that explains why the universe is accelerating and expanding only within 2·$R_s$ radius, that is, the Local Universe.

**5) The Magnetron-Equation (monopole Radio Frequency)**

A specific solution of the unduscular equation for the monopole, 50Khz÷18Ghz, where R is a constant, is

$\frac{R}{r}\Delta H(r,\theta,z,t) = \frac{1}{c^2}\frac{\partial^2 H(r,\theta,z,t)}{\partial t^2}$ (5bis)

Considering that the magnetron is invariant of $\theta$ (z-axis symmetry) in cylindrical coordinates with z-axis symmetry, we are interested in searching for the separable solutions:

f(r,z,t) = $H_3(r)$ $H_4(z)$ u(t) with z-axis symmetry, which satisfy the following four singular conditions for the magnetron;

A) [f(r,z,t) = 0 for r=2R for any t and z] EOWF that can be replaced with $H_3(2R) = 0$;

B) Constraint: squared-amplitude density P(r,z,t) = |ψ|^2 formaalism at (t=0) the force F=f(r,z,0)/(4Pir*z) equals with electromagnetic force & Maxwell's equations.

C) And that verifies the condition that the speed of a photon is the velocity of light, "c."

The function of the z-axis is $H_4(z) = \sin(\frac{E}{\nu h_m}z + \varphi)$

Considering that the magnetron quantum has the same energy as the photon energy, $h_m$, but with a higher value than the Planck constant, E = $h_m\cdot\nu$, where $h_m$ = ~ $5.663714861\cdot10^{-9}$ eVs, the constant of minimum interaction of the magnetron, which is greater than the Planck constant.

We have the wave pulsation of k·z along the axis of propagation, thus. $\frac{E}{\nu h_m} = k$ since $H_4(z) = \sin(\frac{E}{\nu h_m} z + \varphi)$, the wave pulsation is $k = \frac{E}{\nu h_m}$, where E is the magnetron energy and $h_m$ is the magnetron quantum of energy constant.

**6) The microwave monopole with spherical symmetry**

Starting from the unduscular equation particularized for the microwave monopole, 1Ghz÷1.6Thhz we have:

$$\frac{R}{r}\Delta f(r,t) = \frac{1}{c^2}\frac{\partial^2 f(r,t)}{\partial t^2} \quad (1bis)$$

where R = is a constant distance. First, we search for the basis of solutions that have separated variables; therefore, we impose the initial conditions for the tau neutrino, such as to sort out the singular solution of interest:

A) [f(r,t) = 0 for r = 2R for any t] EOWF f(r,t) that can be replaced with g (2R) = 0;

B) Constraint: squared-amplitude density P(r,z,t) = |ψ|^2 formaalism at (t=0) the force F=f(r,z,0)/(4Pir*z) equals with electromagnetic force & Maxwell's equations.

C) and that verifies the condition that the speed of interaction is the velocity of light c.

The above 3 conditions imposed on the Tau Neutrino Effective Unduscular Wave Function f(r,t) uniquely determine the basis of separated solutions of interest.

**7) The Electron Equation**

For the electron, we have an equation that ignores the relativistic mass effect, namely

$$2m\frac{E-U(r)}{h^2}f(r,t) + \frac{R}{r^2}\frac{\partial^2(r\cdot f(r,t))}{\partial r^2} = \frac{1}{V^2}\frac{\partial^2 f(r,t)}{\partial t^2} \quad (11)$$

Where m, E, and V are the electron mass, energy, and velocity in the potential field U(r),

This equation approximates the Schrödinger equation for small radii. That is, for ~ Bohr radius, there is a linear transformation such that, for a certain R, the equation (1) and r<<R, there is a linear transformation that, for the change of variable $r = r_{Bohr}\,\rho$ and

$g(r) = h(\rho)\frac{R-\rho\cdot rb}{R}$ That equivalent to $\frac{1}{\rho}\frac{d^2}{d\rho^2}\left(\rho h(\rho)\right) + \frac{2m}{h^2}\left(E + \frac{e^2}{\rho} - \frac{l(l+1)h^2}{2m\rho^2}\right)h(\rho)$

Thus, the unduscular theory includes all the classical Schrödinger equations and the calculus of the orbital of the atoms with angular dependence, which is the general solution for hydrogen.

Thus, equation (1) is a Quantum Torsion Field Equation of electric, magnetic, light, and gravitational quantum fields for the microscopic world.

We search for the separated solutions $f_{(r,t)}$=g(r)u(t) of this equation that satisfy the following four singular conditions

A) g'(0) = R for r = 0 the initial derivate;

B) The boundary condition g(R)=0. The source note of this paper explicitly specifies the electron field equation,

$$\frac{2m}{\hbar^2}\left(E - \frac{e^2}{4\pi\varepsilon_0 r} + \frac{l(l+1)\hbar^2}{2mr^2}\right)f(r,t) + \frac{R}{r^2}\frac{\partial^2(rf(r,t))}{\partial r^2} = \frac{1}{V^2}\frac{\partial^2 f(r,t)}{\partial t^2},$$

with $R = 3.9228 \times 10^{22}$ m, $f(r,t) = g(r)u(t)$, boundary condition $g(R) = 0$, and requests three concrete things:

1. **verification of the ODE model's values** (relative error against the Bohr/Schrödinger model) for the spectral lines (Balmer, Lyman, Paschen), the Lamb shift, and the hyperfine structure;
2. **comparison of the Taylor expansions** of the two differential equations — Schrödinger and undulatory — such that the free term $r^0$, and the values for $r \lesssim 100\, r_B$, are equal;
3. **detailed explanations**, with plots, of the behavior of the two equations at small values of $r$.

C) And that verifies the condition that the speed of the electron is V= $\frac{\partial f(r,t)}{\partial t}$.

According to Einstein's relativity, the mass is divided by a factor of $\sqrt{1 - \frac{V^2}{c^2}}$

Therefore, we have: $2m\frac{E-U(r)}{h^2\sqrt{1-\frac{V^2}{c^2}}}f(r,t)+\frac{R}{r^2}\frac{\partial^2(r\cdot f(r,t))}{\partial r^2}=\frac{1}{V^2}\frac{\partial^2 f(r,t)}{\partial t^2}$ (12)

***

The electron neutrino is the graviton, and now, in Fermi's theory, we have neutron decay into a proton, electron, and electron neutrino, the same with its antiparticle because it's a Majorana particle: https://uebungen.physik.uni-heidelberg.de/c/image/d/vorlesung/20211/1346/material/Lecture-11.pdf

$n\xrightarrow{\beta-\,decay}p+e+\nu$ the ν electronic neutrino mass according to neutrino oscillation, probably in normal conditions around ν mass ~ 0.00038eV/c$^2$. Thus, the nucleus receives a constant amount of energy from the Sun through τ neutrinos. https://physics.aps.org/articles/v16/13

At the nucleus, the reactions are continuous, and the transformations are as follows, where the abbreviations are: the sign. ⎡ is emission and ⎣ is absorption; posit is the positron, "mag" is the magnetron, and (magnetron is a very light pion π with rest mass 0.007eV/c$^2$), the brackets are for grouping decay:

$$p+\lfloor\nu\xrightarrow{transf}n+posit\lceil\xrightarrow{decay}(p+e+\nu\lceil)+posit=p+(e+posit)=p \quad (23)$$

And τ neutrino will collide with proton, generating the process of continuous migration of pairs of protons-neutrons generating gravity, magnetron, and the μ neutrino that generates the hologram of space and A particle; The A Entanglement appears at any interaction (emission or absorption) of Muon neutrino as an echo of muon neutrino and has the meaning of an Information Field that travels instantly in the hole LU.

$$p+\lfloor\tau\xrightarrow{transf}n+\nu\lceil+A\lceil+\pi^+ +\mu\lceil\xrightarrow{decay}(p+e+\nu\lceil)+\mu\lceil+\lfloor\pi^-+\pi^+=p+(e+\pi^+)=p+\mu+\Delta E \quad (24)$$

The nucleons in the nucleus receive every 62 picoseconds the amount of energy of τ neutrinos, the energy of around ~0.18 eV, from the Sun that generates τ neutrinos. As you can see, the initial proton in an atomic nucleus emits a ν neutrino and a μ neutrino, and absorbs a ν neutrino, so a τ neutrino may also trigger the reaction that results in the initial proton.

And the τ neutrino will collide with a proton, generating the process of continuous migration of pairs of protons-neutrons generating gravity, magnetron for magnetism, and the μ neutrino that generates the hologram of space and A particle; we analysed the Hydrogen atom.

$$p+\lfloor\tau\xrightarrow{transf}n+\nu\lceil+A\lceil+\pi^+ +\mu\lceil\xrightarrow{decay}(p+e+\nu\lceil)+\mu\lceil+\lfloor\pi^-+\pi^+=p+(e+\pi^+)=p+\mu+\Delta E$$

In other heavier atoms, there must exist at least 2 nucleons in a specific amount of time of around 6.22~10$^{-11}$ seconds to interact (the monopole is a very light pion π with rest mass 0.007eV/c$^2$) with pairs of binding spin proton-neutron that change state with each other and exchange magnetron.

*Thus, a τ neutrino, when it interacts with a microparticle, generates a temporal ionization of particles in the initial 3d + T coordinates and emerges after a quantum of time. It behaves as a wave and appears after quanta of time ΔT in the next quanta of space 3d + Δd. The final position is 3d + Δd, & time T + ΔT, continuing the movement from quanta to quanta until the next collision with τ. The space is probably like a sponge with minimum distances between stationary positions available for a microparticle. A particle enters that sponge and exits according to the probabilistic calculus of Onduscular Theory into another place, traveling a space quantum (probably with the restriction that in one position-time of the space-time continuum, there should be only one object: wave, quark, or particle). Also, the wave property of microparticles is given by ν neutrino (graviton), which generates a gravitational ionization (3) with the same mechanism above. The quanta of space in LU (Black Hole excluded) have ~ the dimension of the*

*Quark* ~ $0.43 \cdot 10^{-19}$ m; see the link below: https://arxiv.org/abs/1604.01280. ***This is the wave-particle duality***.
So, 6 informational quarks mean: Mass (M), Magnetic field ($M_g$), Electric charge (E), Gravity (G), Light (L) & Time (T). The informational quarks correspond with the known quarks as follows: Light = Up Quark; Magnetic field = Down Quark; Gravity = Charm Quark; Electric charge = Strange Quark; Time = Top Quark; Mass = Bottom Quark.

| Property:<br>Quark:<br><br>Particle: | Mass<br><br>Bottom | Magnetic Field<br><br>Down | Electric Charge<br><br>Strange | Gravity<br><br>Charm | Light<br><br>Up | Time<br><br>Top | Boundary condition (Dirichlet) Initial Value | Symmetry<br><br>Equation |
|---|---|---|---|---|---|---|---|---|
| Electron<br><br>Plot | 1 | 1838 | -1 | 0 | 0 | 0 | g(R) = 0<br><br>Gaussian distribution's<br><br>$V = \frac{\partial f(r,t)}{\partial t}$ | Spherical<br>$\frac{2m_e}{\hbar^2}\left[E + \frac{e^2}{4\pi\varepsilon_0 r} - \frac{l(l+1)\hbar^2}{2m_e r^2}\right] f(r,t) + \frac{R}{r^2}\frac{\partial^2(r f(r,t))}{\partial r^2} = \frac{1}{V^2}\frac{\partial^2 f(r,t)}{\partial t^2}.$ |
| Anti-electron (positron) | 1 | -1838 | 1 | 0 | 0 | 0 | g(R) =0<br><br>Same as the electron | Spherical<br><br>$\frac{2m_e}{\hbar^2}\left[E - \frac{e^2}{4\pi\varepsilon_0 r} + \frac{l(l+1)\hbar^2}{2m_e r^2}\right] f(r,t) + \frac{R}{r^2}\frac{\partial^2(r f(r,t))}{\partial r^2} = \frac{1}{V^2}\frac{\partial^2 f(r,t)}{\partial t^2}.$ |
| *Neutron | 1838 | -1.913 | 0 | -1 | 0 | 0 | g(R) = 0 | Spherical<br><br>$\frac{2m_n}{h^2}\left[E + \frac{l(l+1)\hbar^2}{2m_n r^2}\right] \cdot f(r,t) + \frac{R}{r^2}\frac{\partial^2(r \cdot f(r,t))}{\partial r^2} = \frac{1}{V^2}\frac{\partial^2 f(r,t)}{\partial t^2}$ |
| Proton | 1836 | 2.793 | 1 | 0 | 0 | 0 | g(2R) = 0 | Spherical<br><br>$m_p\left[E - \frac{e^2}{4\pi\varepsilon_0 r} + \frac{l(l+1)\hbar^2}{2m_e r^2}\right] f(r,t) + \frac{R}{r^2}\frac{\partial^2(r \cdot f(r,t))}{\partial r^2} = \frac{1}{V^2}\frac{\partial^2 f(r,t)}{\partial t^2}$ |
| Magnetron<br><br>(monopole)<br><br>Plot | 0 | -2.793 | 0 | 1 | 1 | 0 | f(2R,z,0) =0<br><br>D[1](f)(0,0,0) = - R &<br>*Dirak quantization*<br>*Gauss law* | Cylindrical $\frac{R}{r}\Delta H(r,z,t) = \frac{1}{c^2}\frac{\partial^2 H(r,z,t)}{\partial t^2}$ |
| Graviton = ν Electronic Neutrino<br><br>Plot | 0 | -1838 | 0 | 1 | 1 | 0 | g(2R) = 0<br><br>$\frac{f(r,0)}{4\pi r^2} \approx$ | Spherical<br><br>$\frac{R}{r}\Delta f(r,t) = \frac{1}{c^2}\frac{\partial^2 f(r,t)}{\partial t^2}$ |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | *Newton's law for r <R/20* | |
| Particle $_{\gamma}$ (Gamma-ray)<br><br>Plot | 0 | 0 | 0 | 1 | 1 | -1 | $g(2R_s) = 0$<br>$V \approx 10^9 \cdot c$ | Spherical<br>$\frac{R}{r}\Delta f(r,t) = \frac{1}{V^2}\frac{\partial^2 f(r,t)}{\partial t^2}$ |
| Particle $E_{(r,t)}$ electric charge | 0 | 0 | 1 | 1 | 0 | 1 | $g(R) = 0$,<br>$g'(0) = R$<br>Maxwell equations | Spherical<br>$\frac{\partial^2 E(r,t)}{\partial r^2} - \frac{\partial^2 E(r,t)}{c^2 \partial t^2} = \frac{1}{Rc}\frac{\partial E(r,t)}{\partial t}$ |
| Photon (Light)<br><br>Plot | 0 | 1 | 0 | 0 | 1 | 1 | $f(R,z,0) = 0$<br><br>Gaussian distribution's | Cylindrical<br>$\frac{R}{r}\Delta H(r,z,t) = \frac{1}{c^2}\frac{\partial^2 H(r,z,t)}{\partial t^2}$ |
| μ Neutrino bound with Higgs Boson<br><br>Plot | 0 | 0 | 0 | -1 | 1 | 1 | $g(2R_s) = 0$<br>$\int_{r_{\min}}^{2R_s} F(r)\,dr = \frac{c}{H_0}$<br><br>$\frac{P(r,0)}{4\pi r^2} \approx$<br><br>*Hubble const.* | Spherical<br>$\frac{R}{r}\Delta f(r,t) = \frac{1}{c^2}\frac{\partial^2 f(r,t)}{\partial t^2}$ |
| Phonon (Audion) The sound particle<br><br>Plot | 0 | -1 | 0 | 1 | 1 | 0 | $g(R) = 0$<br><br>V= 343m/s<br><br>$V = \frac{\partial f(r,t)}{\partial t}$ | Spherical<br>$\frac{R}{r}\Delta f(r,t) = \frac{1}{V^2}\frac{\partial^2 f(r,t)}{\partial t^2}$ |
| Tau τ Neutrino (Time)<br><br>Plot | 0 | 0 | 0 | 1 | 1 | 1 | $f(R,0) = 0$,<br><br>$P(r_0,0) = 0$; $r_0 \approx 11222$ m, Schwarz-schild's radius of the smallest known Black Hole<br><br>$P(R,0) = 0$ | Spherical<br>$\frac{R}{r}\Delta f(r,t) = \frac{1}{c^2}\frac{\partial^2 f(r,t)}{\partial t^2}$ |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Torsion Particle<br><br>$P_T$ | 0 | -1.913 | 0 | 0 | 1 | -1 | f(2R,z,t) = 0<br>$V \approx 10^9 \cdot c$ | Cylindrical<br>$\frac{R}{r}\Delta H(r,z,t) = \frac{1}{V^2}\frac{\partial^2 H(r,z,t)}{\partial t^2}$ |
| Particle $B_{(r,z,t)}$ magnetic charge | 0 | $\pi^+/\pi^- = +/- 3.2910598 \cdot 10^{-16}$ T | 0 | 0 | 1 | 1 | B(2R,z,t) = 0<br>Maxwell equations | Cylindrical<br>$\frac{\partial^2 B(r,z,t)}{\partial r^2} - \frac{\partial^2 B(r,z,t)}{c^2 \partial t^2} = \frac{1}{Rc}\frac{\partial B(r,z,t)}{\partial t}$ |
| *Particle $_M$ (Mass resonance) | 1 | 0 | 1 | 0 | 1 | 0 | g(R) = 0<br><br>Gaussian distribution's<br><br>$V = \frac{\partial f(r,t)}{\partial t}$ | Cylindrical<br>$\frac{2m}{\hbar^2}\left(E - \frac{e^2}{4\pi\varepsilon_0 r} + \frac{l(l+1)\hbar^2}{2m_e r^2}\right)H(r,z,t) + \frac{R}{r}\Delta H(r,z,t) = \frac{1}{V^2}\frac{\partial^2 H(r,z,t)}{\partial t^2}$ |
| *Opto-electron<br><br>Plot | 1 | 0 | -1 | 0 | 1 | 0 | g(R) = 0<br><br>Gaussian distribution's<br><br>$V = \frac{\partial f(r,t)}{\partial t}$ | Cylindrical<br>$\frac{2m_e}{\hbar^2}\left(E + \frac{e^2}{4\pi\varepsilon_0 r} - \frac{l(l+1)\hbar^2}{2m_e r^2}\right)H(r,z,t) + \frac{R}{r}\Delta H(r,z,t) = \frac{1}{V^2}\frac{\partial^2 H(r,z,t)}{\partial t^2}$ |
| *Anti-Neutron | 1838 | 1.913 | 0 | -1 | 0 | 0 | g(R) = 0 | Spherical |
| *Thermal Particle+<br><br>Thermal Particle-<br><br>Plot+<br><br>Plot- | 1 | 2.793 | 0 | 0 | 1 | 0 | g(2R) = 0<br><br>Gaussian distribution's<br><br>$V = \frac{\partial f(r,t)}{\partial t}$<br><br>m = electron mass | Spherical<br>$\frac{2m_e}{\hbar^2}\left[E \pm \frac{l(l+1)\hbar^2}{2m_e r^2}\right]f(r,t) + \frac{R}{r^2}\frac{\partial^2(r\,f(r,t))}{\partial r^2} = \frac{1}{V^2}\frac{\partial^2 f(r,t)}{\partial t^2}$ |
| *Particle $_{En}$ (Physical Entropy Virtual particle)<br><br>Plot | 0 | 1 | 0 | 1 | 0 | 1 | S(2R,0) = 0<br>Gaussian distribution's S(r,t)= r(r)u(t)<br><br>[g'(0) = - R] | Spherical<br><br>$\frac{R}{r}\Delta S(r,t) = \frac{1}{V^2}\frac{\partial^2 S(r,t)}{\partial t^2}$ |

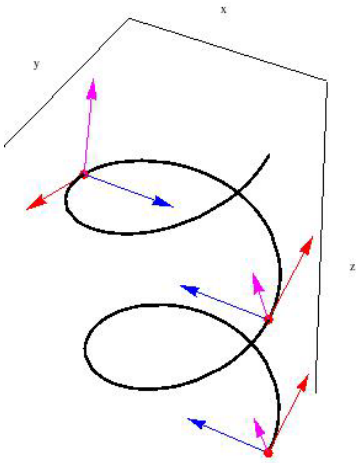
| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Particle $E_x$ | 0 | 0 | 1? | 1 | 0 | -1 | g(R) = 0<br>V >> c | Spherical<br>$\frac{\partial^2 E(r,t)}{\partial r^2} - \frac{\partial^2 E(r,t)}{V^2 \partial t^2} = 0$ |
| Particle S (Synchron) | 0 | 0 | 0 | 1 | -1 | -1 | g(R) = 0<br>f(Rs,t) =0<br>V >> c | Spherical<br>$\frac{R}{r}\Delta f(r,t) = \frac{1}{V^2}\frac{\partial^2 f_1(r,t)}{\partial t^2}$ |
| Activate (Entanglement) | 0 | ? | 0 | 0 | 1 | -1 | H(R,z,t) = 0<br>V >> c | Cylindrical<br>$\frac{R}{r}\Delta H(r,z,t) = \frac{1}{V^2}\frac{\partial^2 H(r,z,t)}{\partial t^2}$ |
| Particle $B_{x(r,z,t)}$ | 0 | $\pi^+/\pi$? | 0 | 0 | 1 | -1 | B(2R,z,t) = 0<br>V >> c | Cylindrical<br>$\frac{\partial^2 B(r,z,t)}{\partial r^2} = \frac{\partial^2 B(r,z,t)}{V^2 \partial t^2}$ |

* Mass resonance Particle $_M$; & opto-electron particles,
Not yet produced or detected: Particle $_{Ex}$; Particle $_{Bx}$; Particle $_{En}$ (Physical Entropy Virtual particle); Part $_{S;}$
* Virtual Entropy Particle $P_{En}$ (as gaps in semiconductors) to induce mass resonance (at Very Low temperature by resonance with Thermal Particle absolute ~ 0 Kelvin = - 273.15 Celsius).

*Thus, we have the postulate that superposition is explained by the temporal ionization of the particle by a tau τ neutrino, so between two positions in space-time is a wave covering a microsphere of interaction in superposition with all states within the sphere of radius $r \approx n \cdot V \cdot T_c$ where $T_c$ is the time quantum of the particle and V is the velocity of particle flux, and n is an integer $\approx 10^3$. The other aspect is the double-slit experiment that emerges from the superposition, which is well explained. Also, the fact that the observer can collapse the wave function in the measurement process is due to the interaction with the particle (usually with light from photons with energy in the visual window around 2-5 eV ), but we interact with the particle with photons of energy less than ~0.9 eV, infrared, depending on the particle (Time quanta). The wave function collapse occurs when interacting, initially in a superposition of several eigenstates, reduced to a single eigenstate due to observation (thus interaction with the external world). The explanation is that observation (if it is in the optical window) generates energetic exchange with the particle, and if it is beyond the threshold value of 1.6295 eV, the gravitational ionization is partially over 1.975 eV or fully blocked temporal ionization. The wave function collapse is within a couple of Time Quanta. There is another aspect of the quantum erasing of the past that refers to the instant traveling of information that goes with a speed V >> c in Synchron Particles, a longitudinal wave that makes the experiment more confusing. The time arrow is only forward, and only with energy support can the time for a set region of space go backward in time. If two particles are entangled, they exchange spin energy; thus, the torsion particle is a longitudinal wave that propagates in a couple of Time Quanta, and the exchange of spin information takes place with V>>c within the Local Universe $V \approx 3 \cdot 10^{17}$ m/s. I guess that the torsion particle associated with wabelining is the same kind of Coriolis force induced by a set of particles with spin pointing in one direction.*

Reintroducing the azimuthal dependence to recover the full 3D photon field.
https://michaelvio.github.io/PhotonAi.pdf
This satisfies the original equation:

$$\frac{R}{r}\Delta H = \frac{1}{c^2}\frac{\partial^2 H}{\partial t^2}$$



Thus, we solve the equation, and we have EOWF; hence, Photon => we divide $H_3(r)/r$ with d unitary of dimension $[L^{-3/2}]$, the final solution approximately of the photon onduscular wave function $H_3(r)H_4(z)\cdot u(t)/r$ of dim $[L^{-3/2}]$ constants as in file https://michaelvio.github.io/PhotonAi.pdf
With $\frac{\partial z}{\partial t} = V_z$ & $\frac{\partial r}{\partial t} = V_r$, and $\frac{\partial z}{\partial r} = 0$; also Pythagoras => $V_z{}^2 + V_r{}^2 = c^2$

Thus, in every moment, we have the speed of a photon in the helicoidal spiral trajectory with the maximum speed that is "c" and components in the radial plane, $V_r$ & $V_z$, in the direction of the propagation.
But the speed $V_z$, in the direction of the propagation, is almost "c" because $V_r$ is small in comparison to the light speed (the variation of the polarization); thus, we approximate $V_z/c \sim 1$

The complete onduscular equation, in the form that reproduces the entire atomic structure of hydrogen, is

$$\boxed{\frac{2m_e}{\hbar^2}\left[E + \frac{e^2}{4\pi\varepsilon_0 r} - \frac{l(l+1)\hbar^2}{2m_e r^2} + \xi(r)\langle\vec{L}\cdot\vec{S}\rangle + V_{HF}(r)\langle\vec{S}_e\cdot\vec{I}_p\rangle\right]f(r,t) + \frac{R}{r^2}\frac{\partial^2(r\,f(r,t))}{\partial r^2} = \frac{1}{V^2}\frac{\partial^2 f(r,t)}{\partial t^2}.} \quad (3)$$

with - $\xi(r) = \frac{1}{2m_e^2c^2\,r}\frac{dV}{dr} = \frac{\hbar\alpha}{2m_e^2c}\frac{1}{r^3}$ for Coulomb $V(r) = -\hbar c\alpha/r$, - $V_{HF}(r) = \frac{2\mu_0}{3}\,g_p\,\mu_B\,\mu_N\,\delta^3(\vec{r})\cdot\frac{4}{\hbar^2}$ — the Fermi electron–proton contact term.

Thus, complete electron Unduscular TauTime, which reproduces the entire atomic structure of hydrogen (Bohr + fine structure + hyperfine) with R preserved as a cosmological constant: $R=3.9228\cdot10^{22}$m, is

$$\boxed{\frac{d^2u}{dr^2} + u(r)\left[\frac{2mE}{\hbar^2} + \frac{r^4}{R^2}\left(\frac{2mE}{\hbar^2} - k^2\right) + \frac{2}{a_0 r} - \frac{l(l+1)}{r^2} + \frac{2m}{\hbar^2}\xi(r)\langle\vec{L}\cdot\vec{S}\rangle + V_{\mathrm{HF}}(r)\langle\vec{S}_e\cdot\vec{I}_p\rangle\right] = 0.}$$

Link: https://michaelvio.github.io/Electron.pdf

https://michaelvio.github.io/ElectronTauTimp.pdf

https://michaelvio.github.io/Electron1s.pdf

Thus, equation (1) is for a massless particle that travels at the speed of light to be reduced to:

$$\frac{R}{r^2}\frac{\partial^2(r\cdot f1(r,t))}{\partial r^2} = \frac{1}{c^2}\frac{\partial^2 f1(r,t)}{\partial t^2} \quad (12)$$

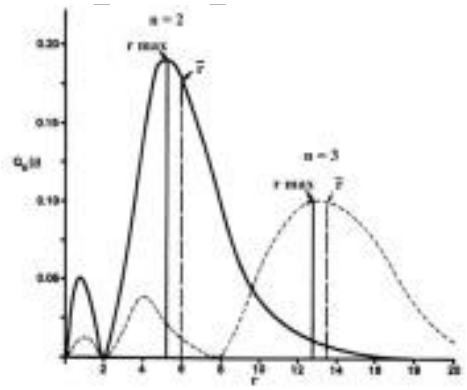
Thus, we denote $f_1(r,t) = r\cdot f(r,t)$ in equation (4) from the article "GRAVITATIONAL EXPERIMENT AND EQUATION" EOWF, that is:

$$\frac{R}{r}\frac{\partial^2(f_1(r,t))}{\partial r^2} = \frac{1}{c^2}\frac{\partial^2 f_1(r,t)}{\partial t^2} \quad (13)$$

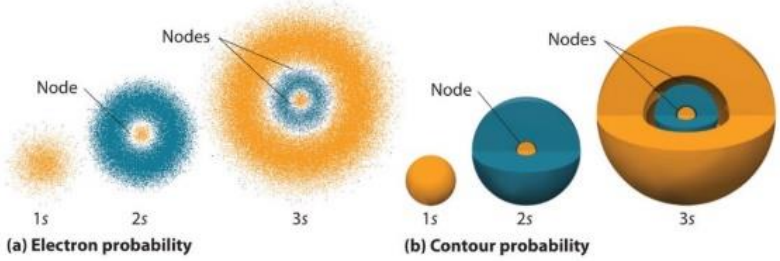
As you can see, the differences are insignificant for the radius Bohr order of $10^{-64}$ m, far less than the precision of $r_b = 5.29241 \cdot 10^{-11}$ m. Thus, equations (15) and (14) with the true coefficient value are similar.
The electron equation is in the file Electron.mw
And the second stage is to demonstrate that equation (16) [which is equation (19.46) in Feynman's article] and (3) are equivalent for a small radius of the stationary state $t_0$, and so that unduscular theory includes all the classical Schrödinger equation and the calculus of the orbital of the atoms with angular dependence, which is f the general solution for hydrogen. $\frac{1}{\rho}\frac{d^2}{d\rho^2}(\rho g(\rho)) + \frac{2m}{\hbar^2}\left(E + \frac{e^2}{\rho} - \frac{l(l+1)}{2m\rho^2}\hbar^2\right) g(\rho)$ (16)
The electron equation, as in Schrödinger, is

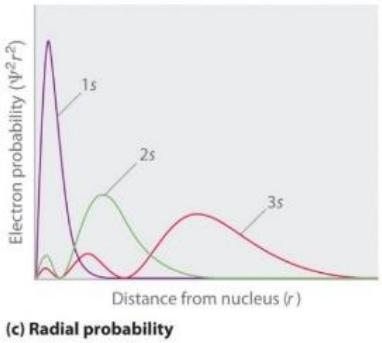
$$f(r) = _C1\, e^{-I\sqrt{E0}\, r}\, \text{KummerM}\left(\frac{4\, I\pi\epsilon + \sqrt{E0}}{\sqrt{E0}}, 2, 2\, I\sqrt{E0}\, r\right) + _C2\, e^{-I\sqrt{E0}\, r}\, \text{KummerU}\left(\frac{4\, I\pi\epsilon + \sqrt{E0}}{\sqrt{E0}}, 2, 2\, I\sqrt{E0}\, r\right)$$

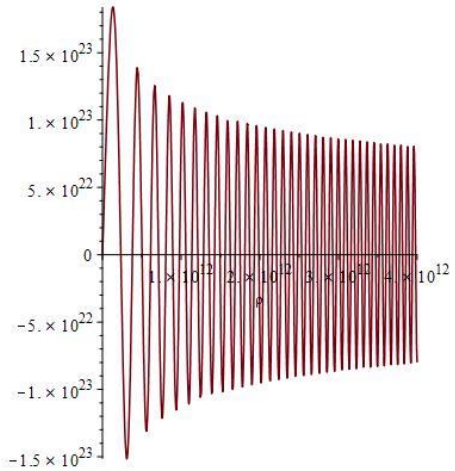
Where _C1 and _C2 are normed constants with the limit condition f (∞) =0.



The electron equation according to the unduscular equation, the value depending on the radius, starting from the unduscular equation without orbital magnetic moment.
And the solution is obtained by solving the equation directly with Maple 21.1 in the file electronPr.mw with time dependency and radial symmetry as below:





Electron Effective Unduscular Wave function of 1 S Hydrogen $E_{Rydberg}$ and velocity V=c/137

**8. The Proton Equation**

For the proton, we have an equation that ignores the relativistic mass effect, namely.

$$2m_n\frac{E-U(r)}{h^2}f(r,t)+\frac{R}{r^2}\frac{\partial^2(r\cdot f(r,t))}{\partial r^2}=\frac{1}{V^2}\frac{\partial^2 f(r,t)}{\partial t^2} \quad (11)$$

Where $m_n$ proton mass (nucleon mass), E is the energy, and V is the velocity in the potential field U(r),

We search for the separated solutions $f_{(r,\,t)} = g(r)u(t)$ of this equation that satisfy the following four singular conditions

A) The boundary condition g (2R) =0 & g'(0)=R.

B) And that verifies the condition that the velocity of the proton is $V=\frac{\partial f(r,t)}{\partial t}$ .

According to Einstein's relativity, the mass is divided by a factor of $\sqrt{1-\frac{V^2}{c^2}}$ Therefore, we have:

$$2mn\frac{E-U(r)}{h^2\sqrt{1-\frac{V^2}{c^2}}}f(r,t)+\frac{R}{r^2}\frac{\partial^2(r\cdot f(r,t))}{\partial r^2}=\frac{1}{V^2}\frac{\partial^2 f(r,t)}{\partial t^2} \quad (12)$$

Thus, the conditions above and the unduscular equation imply that $u(t)=e^{ck_5t}-e^{-ck_5t}$ (13)

**9. The Proton-Equation explicit**

The Quantum Torsion Field Equation should describe all the characteristics of the moving particle in the space coordinate and temporal determination, introducing the Effective Unduscular Wave Function, which is explained in Supposition.pdf, the value of the particle in the space-time at instant t and spherical coords:

$$2mn\frac{E-U(r)}{h^2}f1(r,t)+\frac{R}{r^2}\frac{\partial^2(r\cdot f1(r,t))}{\partial r^2}=\frac{1}{V^2}\frac{\partial^2 f1(r,t)}{\partial t^2} \quad (1)$$

With U(r) = $e^2/r$ and total orbital angular magnetic moment $L^2$ we have

$$\frac{2mn}{h^2}\left(E-\frac{e^2}{r}+\frac{L^2}{2mn\cdot r^2}\right)f1(r,t)+\frac{R}{r^2}\frac{\partial^2(r\cdot f1(r,t))}{\partial r^2}=\frac{1}{V^2}\frac{\partial^2 f1(r,t)}{\partial t^2} \quad (2)$$

Or with $E=E_0\cdot E_R$, where Rydberg energy $E_R$ is: $E_R=\frac{me^4}{32\pi^2\varepsilon^2h^2}$ and $L^2=l\cdot(l+1)\cdot\hbar^2$ azimuthal magnetic number l we have: $\frac{2m_n}{\hbar^2}\left(E_0-\frac{8\cdot\pi\cdot\varepsilon}{r}+\frac{l\cdot(l+1)}{m_nr^2}\hbar^2\right)f1(r,t)+\frac{R}{r^2}\frac{\partial^2(r\cdot f1(r,t))}{\partial r^2}=\frac{1}{V^2}\frac{\partial^2 f1(r,t)}{\partial t^2}$ (3)

And according to the unduscular equation, the value depends on time and radius, starting from the unduscular equation without orbital magnetic moment: $\frac{2m}{h^2}\left(E_0-\frac{8\cdot\pi\cdot\varepsilon}{r}\right)f(r,t)+\frac{R}{r^2}\frac{\partial^2(r\cdot f(r,t))}{\partial r^2}=\frac{1}{V^2}\frac{\partial^2 f(r,t)}{\partial t^2}$

And the solution is obtained by solving the equation directly with Maple 21.1 in the file Proton.mw with time dependence and radial symmetry as below:

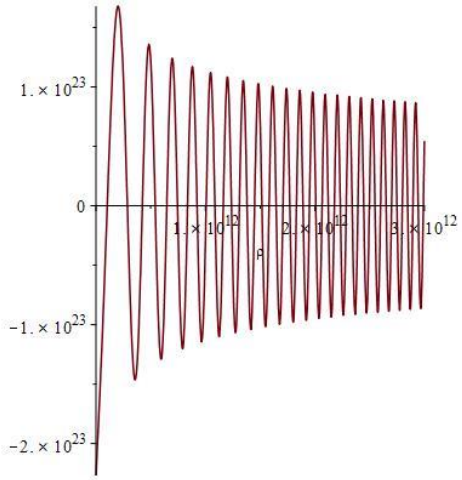
**10. The thermal particle equation:**

Thermal particle equation in spherical coords with spherical symmetry:

$\frac{2m}{h^2}(E+\frac{L^2}{2mr^2})\cdot f(r,t)+\frac{\alpha 1R}{r^2}\frac{\partial^2(r\cdot f(r,t))}{\partial r^2}=\frac{1}{V^2}\frac{\partial^2 f(r,t)}{\partial t^2}$ With R=3.9228·$10^{22}$m length constant and V the speed and m thermal particle mass equal to electron mass + Dirichlet f (2·R, t) =0 & g'(0)=R condition where $\alpha_1$ is

normed thermal diffusivity equal to α1= k/(cs·ρ) where k is thermal conductivity, cs is specific heat capacity, ρ is density and $L^2$ total orbital angular magnetic moment, with no angular magnetic moment, the function is obtained from Electron equation by solving the electron equation with substitute R1=R·α1; The solution neglecting orbital magnetic moment L at speed V= c/137, (c speed of light) equal to the 1S orbital of the Hydrogen atom and energy $E_R$, thus the thermal particle with variable change $f(r,t) = \frac{g(\rho,t)\,(-rb\,\rho + 2R)}{2R}$ as below:

With the plot for t=0 and ρ coordinate, Effective Unduscular Wave function of a Thermal particle with the electron mass, $E_{Rydberg,}$ Velocity α·c:




And plot the function with ρ constant.

**11. Neutron equation:**

The neutron equation in spherical coordinates with spherical symmetry is:

$\frac{2mn}{h^2}(E + \frac{L^2}{2mn\cdot r^2})\cdot f(r,t) + \frac{R}{r^2}\frac{\partial^2(r\cdot f(r,t))}{\partial r^2} = \frac{1}{V^2}\frac{\partial^2 f(r,t)}{\partial t^2}$ With R=3.9228·10$^{22}$m length constant, and V the speed, and $m_n$ as the neutron mass, with total orbital angular magnetic moment $L^2$ and Dirichlet f (R, t) =0 & g'(0)=R conditions. With no angular magnetic moment, the function is:

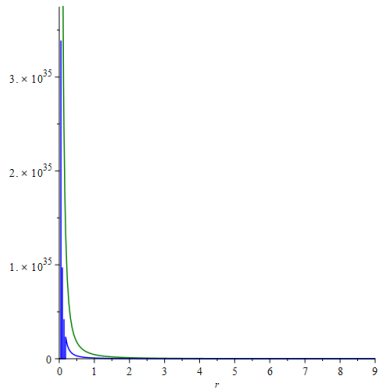
**12. The Phonon (Audion) Equation:**

The phonon particle equation is in spherical coordinates with spherical symmetry:

$\frac{R}{r}\Delta f(r,t) = \frac{1}{V^2}\frac{\partial^2 f(r,t)}{\partial t^2}$ With R=3.9228·10$^{22}$m length constant

With the velocity, "V" = α·c, the speed of the audion (V = sound velocity =>346m/s)in the environment, and the Dirichlet condition is f (R, t) = 0 for constant spatial distance R and g'(0)=R, the effective unduscular wave function of the particle depending on radius r and time t. The solution is like solving the neutrino equation with velocity V = α·c. The phonon (audion) particle equation is on spherical coordinates with spherical symmetry: $\frac{Rb}{r}\Delta f(r,t) = \frac{1}{V^2}\frac{\partial^2 f(r,t)}{\partial t^2}$ With R = 3.9228·10$^{22}$m length constant. With normed propagation constant b, "V" the speed of the phonon in the environment, and the Dirichlet condition is f (R, t) = 0 for constant spatial distance R, and f (r, t) is the effective unduscular wave function of the phonon particle depending on radius r and time t. The solution is like solving the tau neutrino equation with the substitution R1 = R · b, with flux conservation of second order for the phonon in the unduscular equation. The phonon flux equation $\frac{\partial^2 P(r,t)}{V^2\partial t^2} - \frac{\partial^2 P(r,t)}{\partial r^2} = \frac{P(r,t)}{R}$ & $\frac{\partial P(r,t)}{V\partial t} + \frac{\partial P(r,t)}{\partial r} = 0$. With the initial condition P(R,t)=0 and diff(P(0,0),r)=R and V= 346m/s, the speed of sound in air with the phonon flux for null time $t_0$ is $P_{fp}$:

$P_{fp}(r,0) = \frac{\sqrt{2}\cdot R^{3/2}\cdot(e^{\frac{(R-r)\cdot\sqrt{2}}{\sqrt{R}}} - 1)\cdot e^{\frac{\sqrt{2}\cdot r}{2\cdot\sqrt{R}}}}{e^{\sqrt{2\cdot R}+1}}$ $P_{fp}(r,t) = \frac{\sqrt{2}\cdot R^{3/2}\cdot(e^{\frac{(V\cdot t+R-r)\cdot\sqrt{2}}{\sqrt{R}}} - 1)\cdot e^{-\frac{\sqrt{2}\cdot(V\cdot t-r)}{2\cdot\sqrt{R}}}}{e^{\sqrt{2\cdot R}+1}}$




Link https://michaelvio.github.io/FluxAud.pdf

**14. Entropy particle Equation:**

We introduce Effective Entropy, where S(r,t) is the entropy.

$$\frac{R}{r}\Delta S(r,t) = \frac{1}{V^2}\frac{\partial^2 S(r,t)}{\partial t^2} \qquad (2)$$

Thus, equation (2) implies that entropy has a virtual mediating particle that I call the Physical Entropy Particle $P_{En}$ (a virtual particle like an electron hole in a semiconductor) with quarks: Magnetic Field, Gravity & Time.

We understand a virtual mediating particle as something that exists only in statistical physics equations and obeys the Unduscular Equation applicable to all bosons.

Regardless of whether the equations have spherical symmetry or cylindrical symmetry, with null, the rest mass is homogeneous and nonhomogeneous Unduscular Equation, with the velocity being "c", the speed of light, or "V".

Thus, the Entropy particle measures the fatigue state of matter, and it's rather similar to the effect of aging in the case object.

The flux equation for Entropy, considering spherical coordinate symmetry & the speed equal to the speed c, for massless particles. The flux second-order equation is: $\frac{\partial^2 P(r,t)}{c^2 \partial t^2} - \frac{\partial^2 P(r,t)}{\partial r^2} = 0$ (1)

With initial condition P(2R,t) = 0 and diff(P(0,0),r) = -R= $-3.9228\cdot 10^{22}$ Where P(r,t) represent the flux entropy.

Thus the entropy change ~flux/ surface unit = P(r,t) /4πr$^2$ Equation (11):

$$\frac{FEntflux}{4 \cdot \pi \cdot r^2}$$

We seek $P(r,t)$ quadratic in $r$ with a Gaussian envelope at $t = 0$.

**Ansatz at $t = 0$:** $P(r,0) = \mathcal{N}\,(a_0 + a_1 r + a_2 r^2)\exp\left(-\frac{r^2}{2\sigma^2}\right)$

Applying $\partial P/\partial r|_{r=0} = 0$: the coefficient of the linear term at the origin must vanish, so $a_1 = 0$.

Applying $P(2R,0) = 0$: $(a_0 + a_2(2R)^2)\exp\left(-\frac{4R^2}{2\sigma^2}\right) = 0 \Longrightarrow a_0 = -4a_2R^2$

Setting $a_2 = 1$ (normalization absorbed into $\mathcal{N}$): $\boxed{P(r,0) = \mathcal{N}\,(r^2 - 4R^2)\exp\left(-\frac{r^2}{2\sigma^2}\right)}$

(Valid for $0 \le r \le 2R$; negative sign ensures $P \ge 0$ in this domain when $\mathcal{N} < 0$, or we take $|P|$.)

Taking $\sigma = R$ as the natural scale:

$$P(r,0) = \mathcal{N}\,(r^2 - 4R^2)\,e^{-r^2/(2R^2)}$$

The flux equation for Entropy as a virtual particle, considering spherical-coordinate symmetry & the speed equal to the speed of light, is for a massless particle. The second-order wave equation is:

$\frac{\partial^2 P(r,t)}{c^2 \partial t^2} - \frac{\partial^2 P(r,t)}{\partial r^2} = 0$

With initial condition P(2R, t)=0, R= $3.9228\cdot 10^{22}$ m, where P(r,t) represents the flux entropy. Thus, the entropy change ~flux/ surface unit = P(r,t) /4πr$^2$

26 Sep. 2026